

ASHISH SINGH

Machine Learning Engineer | AI Agents | LLMs | NLP

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PROFILE

Machine Learning and AI engineer focused on LLMs, NLP, AI-agent workflows, and production-oriented Python systems. Experienced with transformer-based models, deep learning, data preprocessing, backend APIs, and applied ML research. Published researcher with experience building end-to-end AI applications.

TECHNICAL SKILLS

Programming: Python, SQL, C

AI / ML: LangChain, LangGraph, LLMs, NLP, Transformers, PyTorch, TensorFlow, Scikit-learn, XGBoost

Data: Pandas, NumPy, Matplotlib, Seaborn, Jupyter

Backend / Tools: Django, DRF, PostgreSQL, REST APIs, Docker, Git, Hugging Face, Streamlit, AWS (Basics)

WORK EXPERIENCE

Software Development Intern — Techavatar | Apr 2025 – Jul 2025

- Developed and optimized Python/Django backend APIs supporting data-driven application modules.
- Automated data extraction and transformation workflows using Python and SQL.
- Collaborated with cross-functional teams on application integration, debugging, and workflow improvements.

Salesforce Developer Virtual Intern — Salesforce Internship Program | 2023 | 8 weeks

- Worked with Apex testing and debugging, Lightning Web Components, APIs, Salesforce Flow, and Apex Triggers.

Freelance Machine Learning & Django Developer | Jan 2024 – Present

- Built ML and web application prototypes involving predictive modeling, data preprocessing, Django, PostgreSQL, REST APIs, and interactive interfaces.

SELECTED PROJECTS

Healthcare Claim Denial AI Agent | LangGraph, LangChain, Python, LLMs, Streamlit

- Developed an LLM-powered agent workflow to analyze synthetic healthcare claim data and identify potential denial reasons.
- Designed a multi-step LangGraph workflow for claim analysis, policy lookup, and recommended corrective actions.
- Implemented tool-based retrieval from a local knowledge base of mock payer policies and denial codes with state maintained across workflow steps.

TruthLens — News Credibility Assessment & Summarization | DistilBERT, T5, Transformers

- Developed an NLP application combining transformer-based credibility classification with abstractive summarization.
- Fine-tuned DistilBERT on fake-news datasets and integrated a T5-small summarization pipeline into Streamlit.

VisionVoice — Image Captioning & Audio Generation | CNN-LSTM, Tacotron 2, WaveNet

- Developed an assistive image-captioning application that generates textual descriptions and converts them to audio.
- Applied deep learning techniques for image captioning and speech generation; work published in AIP Conference Proceedings.

PUBLICATION

Ashish Singh et al. "VisionVoice: An Image Captioning Web Application" — AIP Conference Proceedings. DOI: 10.1063/5.0235283

EDUCATION

M.Tech — Artificial Intelligence & Data Science | IIT Patna | 2026 – Present

B.Tech — Computer Science & Engineering | Amity University, Noida | Completed 2025

CERTIFICATIONS

Google Advanced Data Analytics Certificate | Machine Learning A-Z (Udemy) | Django Masterclass — Build 9 Real-World Projects (Udemy)